

Exploring older people's user experiences with VR: A case study of Taiwanese seniors[☆]

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ABSTRACT

Despite much research on virtual reality (VR) techniques, few studies have explored older people's use. Virtual reality is emerging as a promising tool to stimulate physical activity, overcome barriers to aging, and enrich older people's leisure and recreational lives. In Taiwan, medical teams have formally introduced VR equipment to help seniors use serious games for rehabilitation. In the wake of such VR leveraging, our study aims to investigate the operation and usability assessment of VR devices of Taiwan's older people over 70, who felt comfortable in virtual environments throughout the experiment and enjoyed VR task performance and interaction. We used two sets of virtual devices to play game levels and discuss elderly control and usage. The study shows that the two handheld controller designs (symmetric and asymmetric) significantly affected the intuitive mapping of older people's use, with a more intuitive response to the natural mapping of asymmetric controllers mimicking hand shapes. Three key factors (oculomotor disturbance, disorientation, and nausea) were extracted from previous VR motion sickness studies, using the SSQ scale to measure motion sickness issues. The results indicate that more than half of the subjects were not affected by motion sickness issues, indirectly confirming some previous studies that older people are better off than young people with motion sickness. Therefore, more senior-friendly VR content is expected as most older people, physically aging though, yet with fewer sickness issues incurred, are willing to experience VR.

1. Introduction

The World War Generation (1914–1945) is now over 80 years of age, and 2024 is a critical year as all baby boomers (1946–1965) are entering their 60s [1]. Some of these baby boomers are now socio-economic leaders; some are pioneers in developing computer technology; most have used computers in the workplace. Thus, we may expect a significant increase in the demand for technology-based care and social entertainment for older people, as well as for the policies of the aging technology industry.

Internationally, when the proportion of people over 65 years old reaches 7 %, 14 %, and 20 % of the total population, it is called, respectively, an aging society, an elderly society, and an ultra-aging society. Taiwan's National Development Council estimated that Taiwan became an aging society in 1993, turned into an aged society in 2018, and at a relatively fast pace, will soon enter a super-aged society in

2025 [2]; on top of that, the ratio of the elderly population to the total population will continue to increase. Nevertheless, with Taiwan's well-developed information industry and widespread technological development, technological innovation has become critical in improving older people's quality of life and supporting them to live independently. Among others, virtual reality is gaining ground in developing applications for seniors.

Virtual reality (VR) is emerging as a promising medium to promote physical activities among older people, overcome barriers to aging, and increase recreation. Ramalho et al. [3] have encouraged reflections on the multiple impacts of VR on the aging population. VR has evolved from a mere technological novelty to a promising tool for improving the well-being of older adults [4]. VR-based home exercise programs are revolutionizing the physical activity area [5]. Moreover, acceptance of VR among the older demographic is increasing (Ramalho, 2024). In short, immersive and interactive VR experiences offer older people new ways

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to improve physical and mental health and create opportunities for social wellness [6,7]. Practical applications reinforce the potential of VR technology for older adults. For example, In the United States, two companies, RendeVR and MyndVR, are pioneering the development of VR solutions tailored to the needs of seniors. MyndVR [8,9] offers VR headsets and interactive content designed to enhance emotional and mental well-being through virtual travel, reminiscence therapy, and cognitive games. These solutions have been widely adopted by senior living communities across North America, underscoring VR's ability to improve quality of life. Similarly, RendeVR [10,11] focuses on alleviating social isolation by enabling shared virtual experiences, such as revisiting childhood homes or exploring dream destinations, fostering meaningful connections and engagement among older adults in nursing homes. In Switzerland, the unicorn company MindMaze [12,13], a leader in neurotechnology, showcases VR's potential in neuro-rehabilitation. Their platform, used in over 20 countries, integrates neuroscience and VR to aid seniors in recovering from neurological conditions such as strokes and traumatic brain injuries. Through clinically validated systems, MindMaze delivers personalized motor and cognitive training, further demonstrating the transformative role of VR in senior healthcare. In Taiwan, medical teams have adopted VR technology to support senior rehabilitation through serious games. HTC partnered with Ministry of Health and Welfare Taipei Hospital [14] to introduce VIVE VR systems [15], offering art and music therapy for dementia patients. These programs allow seniors to explore Taiwan virtually, relive life stories, and improve their quality of life through interactive rehabilitation.

Thach et al.'s [16] review study pointed out that usability issues and discomfort in using the equipment are major concerns of VR for older adults. Among the usability and discomfort issues, VR controllers and motion sickness are common issues worthy of further investigation. Tuena et al. [17] conducted a systematic review and *meta-analysis* (PRISMA) on the usability of VR clinical systems in aging, screening the full text of 66 articles. The review adhered strictly to PRISMA guidelines. The included studies primarily focused on evaluating the usability, user experience (UX), and feasibility of VR systems for aging populations. Non-immersive VR systems (e.g., desktop-based VR, tablets, and mobile applications) were the most commonly employed. Semi-immersive systems (e.g., large TV or projector screens equipped with sensors for interaction) were also tested in several studies. Fully immersive VR systems (e.g., head-mounted displays or CAVE environments with interaction devices) were less frequently evaluated, with examples limited to studies [17]. Significant rehabilitative effects on various motor and physical measures were observed in all studies assessing the efficacy and usability of VR systems for older adults. Although some technical and interaction-related challenges were reported—such as software bugs, interaction difficulties, realism concerns, and sensor application issues—the included studies demonstrated that the usability of a wide range of VR clinical systems for older adults is generally good, well-accepted, effective, and beneficial [17]. Roberts et al. [18] also showed that older adults viewed both the social entertainment and health promotion of VR positively. Syed-Abdul et al. [19] found that older users viewed VR's usefulness and ease of use as important determinants of the VR experience.

Researchers have frequently noted that game controllers significantly influence the player experience [20]. Akyaman and Alpay argued that the interaction between the player and the game controller should be carefully considered, finding a correlation between the enjoyment of using the controller and the overall enjoyment of the game. Several studies have examined young people using controllers for VR gaming. Tang et al. [21] demonstrated that certain controller movement strategies can significantly reduce motion sickness. Similarly, Li et al. [22] emphasized that intuitive and continuous spatial gestures via handheld controllers can improve VR interaction efficiency, which is particularly valuable with reduced motor agility. Afyouni et al. [23] further revealed that handheld spatial controllers enable more natural

manipulation than touchscreen-based systems, particularly in tasks involving real-world props. Additionally, Polechoński et al. [24] found that lightweight ergonomic modifications—such as the addition of handheld weights—not only enhance physical engagement but also improve enjoyment and comfort. However, research rarely addresses how VR tasks affect the usability of handheld controllers for older adults, despite existing studies on VR's effects, applications, and benefits in social entertainment and healthcare for the senior population. This study seeks to fill that gap by examining the operational challenges and specific impacts on older users, particularly how different controller designs influence their performance, comfort, and perceptions during VR tasks.

Compared with younger adults, seniors perform less well on tasks requiring fine motor control, which may affect their ability and interest in interacting with virtual gaming devices. Some seniors have cognitive disabilities related to the high complexity of virtual devices. Some other seniors are even afraid of damaging the devices if they do not operate them carefully and correctly. Currently, most medical rehabilitation centers in Taiwan use inexpensive virtual devices, mainly handheld controllers, equivalent to human hands in virtual devices. Using handheld controllers to simulate hand movements and interactions, users can perform natural hand movements in a virtual environment, simulating the sensation of touching, grasping, manipulating, and controlling objects in the virtual world to have a more immersive experience. While VR can be an intuitive and natural game control for the youth, there may be a mismatch between actual body movements and corresponding control movements for older people, making it difficult for them to interact with VR games. Questions thus arise: Will seniors be affected by the controllers they use? Is it difficult for them to master the unfamiliar control modes? Besides, controller aesthetics and ergonomics are considered qualitative parts of the gaming experience [25], and the effects of the required control skills of VR gaming devices on elderly users have not been well studied. Thus, it is important to investigate the usability of VR devices for older people from a user perspective.

Another issue accompanying users experiencing VR is motion sickness. Saredakis et al. [26] conducted a comprehensive systematic review of 55 papers based on the PRISMA guidelines for virtual head-mounted display (HMD) literature that met the Simulator Sickness Questionnaire (SSQ) criteria, representing 3,016 subjects (41 % female; age range: 19.5–80; average age: 24, indicating that most participants are young people). The total SSQ means for the older samples were significantly lower than those of the younger samples. However, the older samples here refer to an average age of 35 or above, and these findings are based on a smaller evidence base as a limited number of studies have included older people. Similarly, other research findings [27,28] suggested that motion sickness decreases with age. In contrast, findings from other studies [29,30] indicated that motion sickness increases with age. Regardless of these conflicting results, it is clear that motion sickness affects most users, causing significant discomfort. Moreover, what is in common in the aforementioned studies is the insufficient representation of senior users. Given limited research on VR-associated disorders in people over 70, we recruited Taiwanese seniors over the age of 70 to conduct motion sickness tests and study VR gameplay sickness issues in older people. Our objective is to determine whether dizziness increases with age, making VR less suitable for older people, as commonly believed, or whether dizziness actually decreases with age, presenting a different perspective for VR design.

Although comparative studies of VR user experience are gaining attention, studies on older users still need to be more extensive. In recent years, studies have proved that VR interactive experiences are indeed beneficial to older people's physical and mental health, and the industry has gradually developed content suitable for them. Therefore, our study tested the control of hand operation and motion sickness of older people using VR devices with handheld controllers. The findings would contribute to the senior user experience of virtual devices in social entertainment and health promotion.

2. Related work

2.1. The elderly related issues

The globe has gradually entered an aging world. Due to medical advances and modern people valuing health, many older adults over 70 still walk and move freely. In the meantime, video gaming has become a mainstream activity. Gaming is no longer limited to single geeks; players are the general population, spreading to the elderly, married, female, minority, educated, and middle-income groups [31]. Older co-players express higher levels of positive emotions and happiness, companionship, and emotional support for video games [32]. Most older people in contemporary society own smartphones, and the number of them playing mobile games is growing. Some also use physical game machines for self-fitness, friend interaction, and parent-child entertainment. The use of Nintendo Wii as a sports game platform by older people has exerted positive effects, such as physical function improvement, depression reduction, cognitive stimulation, social support, and enjoyment [33]. In addition, VR interactive content is gradually entering the family market, which is suitable for parents, guardians, children, and even older people to participate together. Indeed, VR devices allow for immersive exploration of the 3D world, and VR interactive experiences train the body, sensory integration, and coordination. Although most games available today are for young people, particularly VR games, content for senior citizens is being introduced slowly, offering more options for recreational activities.

Walking has been shown to stimulate executive function in sedentary older people [34]. Shafir et al. demonstrated that exercise movement can modulate and enhance one's experienced emotions, finding further that exercise characterized by stretching the body is liberating and leads to higher positive emotional well-being [35]. VR games corresponding to such study findings are being created to offer immersive experiences for healthy entertainment. Benoit et al. [36] showed that using VR can stimulate recollections of autobiographical memory in 18 healthy older individuals (average age 68.2). According to Kim (2017) et al., older adults with mild to moderate Parkinson's disease could perform walking exercises on a slow treadmill for 20 min per 5-minute round using immersive VR. The visual scenes in the simulation were designed to match the older adults' motor behavior, and the simulated environment's panning speed matched the participants' walking speed.

The VR experience has a unique immersive effect; however, some potential issues accompany the use of VR, such as handheld controller operation and motion sickness issues [37]. In particular, VR is challenging when it comes to older users. Older people might exhibit greater reluctance or resistance towards adopting innovative technologies compared with their younger counterparts, primarily as a result of variations in cultural influences, educational backgrounds, and life experiences [38]. Most digital products are complex to operate. Some users (novices or older people) often stop using certain features after several failed attempts [39]. Older people have cognitive barriers to the complex and difficult-to-manipulate devices in high-tech products, and some of them even refuse to use the devices for fear of damaging them by inadvertent operation. This situation emphasizes the importance of conducting senior user experience research on emerging and innovative technologies [40]. In addition, seniors have lower performance on tasks requiring fine motor control compared with young people. This requirement may affect their ability and interest in interacting with gaming devices. When using VR devices, users input information through hand controllers and receive feedback from VR displays. VR products have many functions and are often complex to operate. Most novices (especially older people) will not be able to use certain features without some degree of trial-and-error learning or reference experience.

2.2. VR serious games for older adults in Taiwan

Taiwan is facing challenges as it moves towards becoming a "super-

aged society" by 2025, with over 20 % of its population being aged 65 and above. These challenges encompass a changing demographic structure, labor shortages, and ensuring that the elderly population lives longer and healthier. Nonetheless, Taiwan's solid technological base and innovation capabilities put it in a favorable position to develop high-tech products and services for seniors. Notably, Taiwanese companies have demonstrated impressive achievements in manufacturing VR hardware, developing software, and creating content. The advancement of VR technology in Taiwan presents excellent opportunities to enhance the quality of life, healthcare, and social entertainment for seniors, positively impacting the challenges of an aging society.

Utilizing XR technologies can help optimize the lives of older people and allow them to experience a simulated world or objects without the need for physical movement, providing them access to a world of engaging and therapeutic virtual experiences. Established in July 2020, The Yue Xin Community Long-term Care Center, specializing in care for individuals with dementia, located at Taipei Hospital and under the Ministry of Health and Welfare, recently launched an innovative program using VR technology. This program allows seniors to virtually tour Taiwan comfortably, mimicking a business-class seating experience without the need for physical travel. By integrating VR technology into its therapeutic activities, it enhances reminiscence and sensory stimulation. It is the first of its kind in Taiwan to apply VR in delaying dementia symptoms (Ministry of Health and Welfare Taipei [14]).

Syed-Abdul et al. [19] studied 30 Taiwanese senior participants between the ages of 60 and 95, with each senior participant engaging in a 15-minute VR experiential activity twice a week for six weeks. The focus of the study was to assess the use of nine VR applications that motivate seniors, promote physical activity, and provide entertainment for these seniors. The study's results showed that the older participants had overall positive experiences with adopting the new technology. This positivity was broadly consistent with their perceiving VR to be helpful, easy to use, and enjoyable, indicating their positive attitudes towards adopting this new technology. Studies have also found a significant positive relationship between desire for new experiences (seeking experiences), quality of experience, and value of experience, and this value of experience provides empirical support for the conceptualization of VR leisure activities for older adults. Jeng et al. [41] reviewed the behavioral intentions of Taiwan's senior participants experiencing VR leisure activities and proposed a new experience model with the value of experience as a mediating variable. The study collected valid samples from 294 Taiwanese seniors who continuously participated in the activities for ten weeks, and the results showed that the VR experience value of seniors could effectively predict behavioral intentions (including purchase and continuous participation) and acted as a mediating variable between experience quality and behavioral intention. Both desires for new experiences and experience quality were significantly and positively associated with experience value, which in turn influenced behavioral intentions and provided empirical support for the conceptualization of VR leisure activities for seniors.

Cheng et al. [42] developed a program that combined VR and aromatherapy to enhance the psychological health of older people residing in long-term care facilities. The program involved a nine-week course, with each session lasting two hours. The participants were immersed in multisensory experiences, such as meditation in a virtual forest with sandalwood scent, which stimulated and relaxed them visually, auditorily, and olfactorily. The study found significant improvements in all targeted psychological health outcomes, with more notable benefits for participants aged 80 and above. According to Lin et al. [43], the combination of 3D virtual reality and horticultural therapy has significantly improved institutionalized older people's physical and mental health. The study, conducted over nine weeks, showed that participants in the experimental group experienced enhanced health status, increased meaning in life, greater perceived mattering, and reduced levels of depression and loneliness. Furthermore, the benefits persisted for two months after the program ended.

These findings suggest that such programs could be effectively implemented on a larger scale in long-term care facilities.

The aforementioned authors have highlighted the positive reception and potential advantages of VR technology in improving senior citizens' quality of life and leisure experiences. Their studies emphasized the importance of experience value in influencing the attitudes and intentions of older people to engage in VR activities. Taiwanese seniors also demonstrated an interest in and enjoyment of VR experiences as a form of entertainment.

2.3. VR device operation status

Virtual reality technology has been around for decades, but it wasn't until recent advances in mass-market head-mounted displays (HMDs) and Meta's substantial investment in the virtual gaming market that interest in VR was rekindled and gained widespread attention. Current VR systems, such as Oculus Rift or HTC VIVE, combine next-generation motion controllers with 360-degree HMDs to handle users' hand and head movements [44]. The user interface and 3D interaction technologies are critical to virtual reality in an immersive virtual environment. At the same time, head and hand tracking are typically enabled by 6 DoF tracking technologies, which allow the three-dimensional display of virtual objects in the virtual world to interact naturally and visually with the user. As a result, users can use handheld controllers to grab or select virtual objects in the virtual world or manipulate them within their reach.

Recent market trends show that VR devices are becoming popular and affordable among consumers. As thousands of VR software have been offered on digital platforms such as Steam, Viveport, Oculus Store, and Google Play, many users are more willing to try various VR interactive applications [45]. With the emergence of affordable VR devices such as the Meta Quest series, many productions and projects are using virtual environments for applications related to digital entertainment (3D games and 3D movies), digital architecture, simulated educational training (e.g., flight simulation, driving simulation, and surgical simulation training), and remote manipulation (e.g., robots). These applications and virtual environments provide users with immersive and enjoyable experiences through interacting with the virtual world. For example, Reer et al., by experimenting with Self-Determination Theory, demonstrated that the VR version of the game was more enjoyable than the non-VR version [44]. In addition, the VR market is becoming more active due to the Covid-19 pandemic. It was estimated that the global VR market would reach US\$15.81 billion in 2020 and grow at about 30 % per year [46]. More recent studies showed that the global Augmented and virtual reality market size was valued at US\$ 34.53 billion in 2021 and was expected to expand at a CAGR of 38.69 % during the forecast period, reaching US\$ 245.75 billion by 2027 [47].

Selecting a target in virtual reality is one of the basic operations that help the user to get the desired target for the following action. Usually, users need to manipulate multiple targets in a specific order to complete tasks such as options or actions. In this case, target selection and action movement are required, and multiple desired targets are continuously manipulated to enable immersion in a continuous virtual experience. In virtual reality based on affordable head-mounted displays and controllers, head gaze, handheld controllers, or unaided aerial gesture input are typical interaction methods [48]. Caggianese et al. [49] demonstrated that handheld controller interactions are more responsive regarding current techniques than unaided aerial gesture interactions. De Paolis and De Luca [50] showed similar results in a VR human organ exploration and navigation experiment, where users reported that the HTC VIVE handheld controller was better than the Myo gesture control arm. Indeed, more expensive and advanced Leap Motion controllers for real-time skeletal tracking of hands and fingers also exist, which can easily enable controller-free aerial interaction in virtual reality.

Although there is no definitive conclusion as to which VR controller is the best, the handheld controllers are currently the most stable choice,

taking into account user experience and technology threshold. Therefore, most VR devices are equipped with handheld controllers. In terms of market share, there are two trends: the symmetric design of the VIVE series and the asymmetric design of the Quest. In 2016, as a pioneering consumer-level VR device, the VIVE, co-developed by Valve Corp. and HTC, adopted a symmetric design of handheld controllers, similar to the Playstation Move launched by Sony in 2010. The most significant feature of the VIVE controller is its two touchpads, following in the footsteps of the Steam Controller that came with Valve's Steam Machine. This design aimed to combine a gaming controller with a mouse-and-key experience. Even though the partnership with Valve only lasted for the first VIVE, the controller continues to appear in the VIVE series, further in the VIVE Pro 2, launched in 2023.

The other mainstream controller design, the Quest option, is wholly inherited from console controllers, with the thumbsticks, triggers, and buttons found on traditional home console controllers. This design is intended to reduce the learning curve for the user at the entry-level. The content of VR games is designed to provide a relatively immersive virtual environment with less complex controls such as buttons or triggers [51]. Chirico et al. [52] analyzed the performance of a 91-year-old woman (Tina) in both reality and immersive VR situations and further evaluated two different VR interfaces: controllers and sensors. The results show that performance, stress, and subjective reports were similar across other conditions, with both VR interfaces resulting in positive ratings and no motion sickness.

It is worth noting that eye-tracking [53] and motion capture [54] are counted as more intuitive interaction solutions. However, considering cost and system stability, handheld controllers remain the most practical choice for current device manufacturers. Accordingly, considering the universality and affordability of the devices, our study focuses on the experience and evaluation of affordable VR devices for older people and thus selected affordable VR devices, namely Meta Quest 2 and HTC VIVE.

2.4. Handheld controller operation modes

Handheld controllers are the most feasible alternative to hand manipulation in virtual reality. Although handheld controllers cannot be as agile as human hands, they compensate for this imperfection through several control methods. For instance, the trigger and touchpad on the HTC VIVE controller [55] or the trigger and thumbstick on the Meta Quest controller [56] allow users to control movement by simply moving their thumb and index finger along its surface and interact with the object in a virtual environment. Handheld controllers also provide multiple buttons or joysticks for selecting and operating task controllers controlling interface operations to allow users to use these features to achieve advanced types of interactions. Such controllers offer the distinct advantage of realizing such complex real-world interactions as grabbing and pointing in a simple way.

According to Bowman's 3D user interface interaction theory, common interactive tasks include navigation (travel and wayfinding), selection/manipulation, system control, and symbol input [57]. Based on this theory and the following studies of Caputo et al. (2015), Sun et al. [58] further classified four key factors for the UI design in VR applications: navigation modes, operation methods (selection/manipulation), observation scales (system control), and background options. However, the fourth factor is a system control subtask, so our study focuses mainly on the three key factors of interactive tasks. The navigation tasks include path movement and roaming; selection/manipulation tasks include selection, movement, and rotation; and system control includes commands that request the system to perform specific functions that change the system interaction mode or system state.

The first interactive task, "navigation," includes travel and wayfinding. Our study focuses rather on travel or displacement; the task of finding the target and direction cognitive test is not considered for the present study. VR displacement technology mainly includes the

following modes: real walking, redirected walking, walking-in-place, teleporting, joystick, swimming, arm-swinging, etc. [59]. Due to the constraints of the base station's range and the room's actual size, a user's movement in the virtual environment can quickly reach its limits. In such cases, one must rely on the aforementioned motion techniques to move around. The most popular among these are artificial locomotion and teleportation. Artificial locomotion allows for stable and continuous movement within the VR space based on controller input (Directional controllability, such as a joystick or touchpad). Teleportation, on the other hand, also relies on controller input. Users select a target location by pressing a button and aiming the controller at the desired spot. Upon releasing the pressed button, the user is instantly transported to the chosen location [60].

The second interactive task, "selection/manipulation," mainly refers to the selection/grabbing mode (Trigger controllability) in VR interaction. In the 3D VR virtual environment, many interactive objects simulating real-world physical effects are incorporated to enhance immersion and realism. Users first touch the object they wish to interact with using the controller. At this point, the object will highlight its outline, indicating that the user has selected it. They then press down on the trigger to virtually grasp it. The target object will move, rotate, etc., in real time, corresponding to the controller's movement. When the trigger is released, the object is let go and will fall according to inertia [58]. Apart from the trigger, most handheld controllers have additional buttons to simulate more complex object interactions or hand movements. For example, the VIVE controller places a grip button on each side of the handle. Users can press these buttons to 'hold' objects (such as bows or guns) without continuously pressing the trigger, enabling more nuanced interactions (like shooting by pulling the trigger while holding a gun). On the other hand, the Meta Quest series features a second trigger on the inner side of the controller, simulating the movement of the middle finger. This setup allows for the simulation of actions like making a fist (by pressing both the trigger and grip button), pointing (by pressing only the grip button), or opening the hand flat (pressing neither).

The third interactive task, "system control": Currently, VR uses a touchpad/thumbstick and buttons to give choice and control, allowing users to adjust the perspective, direction, and movement further, as well as to select different options from virtual panels [57]. In immersive experiences, the comfort level can vary due to individual differences among users, such as tolerance to motion sickness, eye spacing, height, and arm span. VR systems offer simple adjustments, like turning in place, to minimize discomfort during use. Beyond that, there are also relatively complex UI interactions. Unlike traditional digital experiences, these 2D interfaces, once summoned, are generated before the user and hover in the virtual environment, akin to holographic projections. With handheld controllers, there are two ways to interact. First, using a laser pointer generated at the front of the controller to aim at the interface and then pulling the trigger to select is similar to using a remote control. Second, by directly touching the UI elements with virtual hands. 2D UIs are commonly used for system control and game menus. System controls are primarily accessed via the system button (Functional controllability) on the handheld controller and can adjust the overall VR experience, such as system settings or ending the current experience. Furthermore, most VR experiences also feature game menus to assist with adjustments specific to that experience, such as volume control or accessing files [61].

Natural mapping serves as the cornerstone for achieving stimulus-response compatibility within the specialized realms of ergonomics and human factors. The escalating complexity of human-computer systems and the demand for multi-task information processing present challenges and garner attention. Designers establish mappings of the spatial relationship between buttons and displayed content in the user interface design of an electronic device, so users can operate it easily and efficiently [39]. Tsang and Chan [62] investigated spatial stimulus-response compatibility effect on task performance in the dual-task

condition with different display-control position mappings. Conducting a usability study on wearable fitness devices with diverse physical designs to evaluate how mappings between the features of these devices align with users' perceptions of fitness technology [40]. Human-machine systems for multi-task information processing, especially in tasks involving response compatibility of visual, auditory and tactile spatial stimuli, which more require activities to be performed under different spatial stimulus-response mapping conditions [63]. Mapping in game controllers refers to the intuitive design of game controllers that align closely with real-world actions. For instance, a steering wheel controller for racing games mimics actual driving. The mapping design of video game controllers, encompassing ergonomics, button layout, and feedback, significantly influences player interaction and overall enjoyment of the game [20]. Game controllers that are both functionally effective and emotionally satisfying tend to enhance the gaming experience.

Virtual reality is typical multi-tasking information processing human-machine system. In the specific context of this study, "mapping" denotes the relationship between a handheld controller and VR appliances. When designers establish natural mapping for the relationship between handheld controllers and the contents of VR appliances in the user interface design of VR devices, users can seamlessly and efficiently interact with the system. According to Norman [64], natural mapping contributes significantly to users' intuitive and immediate control-response of system information. While the principles of mapping are inherently straightforward [64], their application becomes notably challenging in the design of VR appliances. This complexity arises from the multitude of functions embedded in VR appliances, presenting a formidable task for designers aiming to create a natural mapping for the user interface within the confined space of the handheld controller. The overall handheld controller operation can be addressed by examining cognitive skills and then showing how these skills "map" to game performance to deal with the degree of natural mapping instilled by the handheld controller. This is a way to show how "natural" it may be to use a particular controller in a game [25]. When natural mapping is thoughtfully integrated into the interface design of products, users, particularly the elderly, stand to gain numerous advantages. These include accelerated learning, quicker reaction times, reduced error rates, and heightened user satisfaction [65].

2.5. VR sickness

In VR experiences, not only do users experience 3D immersion, but sometimes it may be accompanied by symptoms of motion sickness, called simulator sickness or cybersickness, also explicitly called VR sickness [66]. Besides dizziness, VR sickness's main symptoms may be eye fatigue, disorientation, and nausea [67]. New HMD-VR technologies have vastly improved visual and tracking resolution, yet current literature suggests that motion sickness still occurs to some extent in the player's body [66,67]. Furthermore, using VR, even without players' interaction or movement, can incur sickness. Allen et al. [68] tested 73 participants using Oculus Rift to view just 3D stereoscopic movies. The study found that VR sickness levels varied, and individual differences in sensitivity to dynamic visual depth cues correlated with reported discomfort levels. Therefore, VR sickness as a significant problem has forced VR hardware and content developers to study and solve it.

Space-based VR interactive, immersive experience simulating movements relies on displacement technology to alleviate motion sickness symptoms. The aforementioned first VR interactive task – navigation – depends on locomotion techniques, and the VR locomotion mode critically affects motion sickness. However, most studies focus on VR locomotion techniques, with relatively few studies on user experience [59,69]. Although user experience studies have been conducted to compare different controller locomotion modes in recent years, hardware remains the target of comparison [70,71]. According to studies of VR locomotion techniques, teleportation mode was eventually dominant

as it seemed to be the best mode to avoid motion sickness [72]. Besides, empirical user studies [70] reconfirmed that teleportation mode controllers made users feel the slightest motion sickness. According to the related literature on motion sickness, mostly contributes to optimizing VR hardware, teleportation mode controllers cause the slightest motion sickness. Moreover, although more recently, *meta*-studies of dizziness symptoms have been conducted via user-centered, HMD-mediated analyses of VR content [26], and comparative studies of VR user experiences are gaining attention, studies on older users are relatively limited. And recent studies have confirmed that interactive VR experiences contribute to older people's physical or mental health, and VR content for older people is being developed. Therefore, our study tested the motion sickness of older users using VR devices with the teleportation mode controller.

Regarding users' motion sickness, the existing literature presents disparate findings concerning the increase or decrease with age. In Häkkinen et al.'s study (2002), subjects aged 18 to 41 wore HMDs to experience a virtual race environment, and the results showed a significant increase in motion sickness scores in the older age group compared with the younger age group. Park et al. [30] found a similar age difference, where the percentage of dropouts in the car simulation experiment was higher in the older age group (70–90 years old) than in the younger age group (21–50 years old). In contrast, the finding of Saredakis et al. [26] suggested the opposite, although the age group distinction is relatively young: participants with an average age below 35 reported higher total motion sickness score compared to those with an average age above 35. Nevertheless, this finding implied that VR motion sickness becomes less and less evident with age growth. Paillard et al.'s research finding (2013) corresponded with a similar hypothesis that susceptibility to VR motion sickness may decrease with age. 167 healthy subjects, consisting of 84 women and 83 men with an average age of 52.9 (± 19.2), participated in this study without any history of vertigo, dizziness, or severe neurological conditions. The research results showed that the MSS (motion sickness susceptibility) of healthy elderly individuals decreases with age. This finding suggested that sensitivity to motion sickness diminishes as age increases. Similarly, Kim et al.'s study (2017) presented a similar result, specifically on seniors with Parkinson's disease, who did not elicit VR-associated disorders. The finding suggested a particular phenomenon that VR-related disorders did not affect seniors with Parkinson's disease. The authors explained that they did not measure possible deficits in sensory integration for the participants. Therefore, seniors with Parkinson's disease may not have sufficient deficits in sensory integration to generate simulator sickness via sensory conflict.

If we examine closely, the above former group that suggest motion sickness increasing with age indeed involves VR experiences with more intense actions such as racing and car simulation, which can be supplementary factors for dizziness increase. On the other hand, the latter group conducted experiments on older people experiencing milder VR interaction, finding that over half of the senior participants experienced no significant motion sickness, and their symptoms were milder compared to younger populations reported in prior research. Our study aligns more with the latter group. We explore the hypothesis that age-related changes in sensory integration may explain why older adults experience less severe motion sickness than younger individuals. This challenges conventional beliefs and emphasizes the need for further investigation. It is worth noting that the identification of older age in the latter group varies from an average of 35 years old, middle-aged to older adults. What is in common is that the participants were all under 60 – almost no research focused on the very elderly (those over 70 years old). Therefore, we aim to address gaps in existing research, such as limited data on motion sickness in seniors over 70, the effects of specific VR tasks on older populations, and how the usability of controllers may help mitigate motion sickness.

As the super-aged era approaches, we will explore senior citizens aged 70 and above using VR devices. We opted for including pre- and

post-experiment SSQ assessments and focused on short exposure durations (10-minute intervals) to capture immediate effects. This approach aimed to provide a nuanced understanding of motion sickness in older adults and contribute to addressing such research gaps. Our study attempts to test seniors' relative resilience to motion sickness. This resilience could probably make them more suitable for VR applications, particularly rehabilitation and health-related interventions.

2.6. The simulator sickness questionnaire (SSQ)

The Simulator Sickness Questionnaire (SSQ) [73] is now widely used to assess the occurrence, nature, and severity of motion sickness symptoms induced in VR environments. Kennedy et al. published the Simulator Sickness Questionnaire (SSQ) in 1993, initially developed for military flight simulators. Bimberg et al. [74] described the challenges inherent in applying SSQ to virtual reality research, highlighting the significant discrepancies in related work. The study also suggests how to improve the representativeness of SSQ results in future studies and encourages researchers to investigate symptoms of motion sickness caused by VR environments with simpler measurements. Because our study subjects are aged 70 and above, we use the Kennedy 1993 and Bimberg 2020 Simulator Sickness Questionnaire (SSQ) with its weighted calculation. The SSQ consists of 16 items rated on a scale of 0 to 3 (the value is based on the degree to which a particular symptom affects the patient: 0 = none; 1 = mild; 2 = moderate; 3 = severe). Three symptom clusters were assessed by 16 signs: oculomotor disturbance (e.g., blurred vision), disorientation (e.g., dizziness), and nausea (e.g., vomiting) (Appendix). There are different weightings based on the three symptom clusters, with higher scores indicating higher symptom status [74,75]. Given the elderly participants, it was not possible to confirm whether they were healthy and had no motion sickness. The zero-baseline principle was not used in this experiment; we thus adopted a pre- and post-experimental questionnaire Brown et al. [76] suggested.

3. Methodology

3.1. Experiment participants

A total of 35 older adults (17 males, 18 females, aged 70–85, average age 68.2) participated in the study, each receiving a 1000 NTD incentive. Participants met these criteria: able to walk and climb stairs independently, no diagnosed balance issues, and no prior VR experience. To ensure consistent familiarity, all participants attended a standardized training session on using VR equipment and handheld controllers. Before the experiment, they completed a pre-test questionnaire covering demographics, prior motion sickness, and physical condition (16-item SSQ). During training, an observer provided one-on-one guidance to ensure participants could operate the system confidently. Participants were instructed to stop right away if they felt dizzy or unwell. Throughout the experiment, they remained standing and were closely monitored by an observer to ensure safety.

We use the Simulator Sickness Questionnaire (SSQ) as it remains a widely recognized tool for assessing motion sickness in VR research, particularly for its established reliability and validity across diverse studies. To mitigate potential biases and ensure consistency in responses, we provided the following measures:

Standardized Instructions: Participants received clear verbal and written explanations with examples before the experiment.

Observer Assistance: Trained observers assisted participants with any questions during the process.

Uniform Administration: Questionnaires were completed immediately after each session under the same conditions to minimize external influences.

3.2. Equipment and experimental procedures

In order to apply to elderly entertainment scenarios, our research experiment considered the development and maintenance costs of the current mainstream VR platforms and further divided the common handheld controllers in the market into symmetric and asymmetric types for usability studies. The symmetric controller does not distinguish between left and right gripping types, and the button design and touchpad (directional control) are symmetrical as well. Asymmetric controllers have different curves for left and right gripping types to meet the gripping suitability; also, the button design and thumbstick (directional control) are designed according to the positioning of fingers and palm.

Concerning the VR hardware, our experiment selected Meta Quest 2 and HTC Vive controllers (Fig. 1), boasting the best immersion effect and sufficient freedom of virtual operation. The symmetric type is represented by the handheld controllers of HTC Vive, developed by Valve for VR experience in 2016 and being used to date. The asymmetric type is represented by the Meta Quest 2 handheld controllers, the button configuration similar to that of the previous home game console, with buttons A and B on the left controller and X and Y on the right, plus a thumbstick on each. This design is more attractive to the young gaming community.

Each senior participant was randomly assigned to use one of the VR devices (Meta Quest 2 or HTC Vive) to play a game for 10 min. After removing the HMD, they filled out questionnaires on controller usability and motion sickness (SSQ), followed by a 10-minute recovery. Then, they used the other VR device for the same task and completed the same post-test process. Each participant thus experienced both devices. The 10-minute intervals were based on prior research indicating that VR-induced sickness tends to increase after this duration [77,78].

3.3. Experiment: adopted VR video games

Since participants were seniors aged 70 + with little to no gaming experience, we chose *The Lab* – Postcards (introductory level), a simple VR game with easy-to-understand rules. The game allows players to explore scenic 3D environments and interact with objects using a controller. Before gameplay, an observer explained the game, assisted with equipment setup, and guided participants through basic actions in the virtual environment.

(I) Direct interaction with the robot dog (Entry Level)

I.1. Locate the robot dog and get familiar with the virtual environment.

I.2. Move toward the dog using navigation controls.

I.3. Bend or squat to interact with the dog using the controller (e.g., patting).

(II) Indirect interaction using a stick (Advanced Level)

II.1. Find and locate the stick in the environment.

II.2. Move toward the stick using virtual navigation.

II.3. Bend or squat to pick it up with the controller (trigger to grab).

II.4. Throw the stick by moving your arm and releasing the trigger.

II.5. Watch the robot dog run toward the stick.

II.6. Move near the dog again using navigation.

II.7. Retrieve the stick from the dog's mouth (bend or squat, grab).

II.8. Repeat throwing and retrieving steps (4 to 9) as before.

During the task, participants used the controller to navigate, select, and perform actions using natural mapping. These interactions are outlined in Table 1.

3.4. Post-experimental survey

After using Meta Quest 2 and HTC Vive (order randomized), participants completed a usability questionnaire evaluating handheld controllers on a 10-point scale. Items included trigger, directional, functional, mapping controllability, comfort, and enjoyment. A 4-point SSQ motion sickness survey was also administered before and after each VR session to assess symptoms such as dizziness and nausea. The post-test surveys took 7–10 min, serving as a recovery period. These questionnaires aimed to understand older adults' hand interaction, VR adaptation, and motion sickness responses.

4. Statistics and analysis

4.1. Handheld controller analysis

After using both devices, participants rated the usability of symmetric and asymmetric handheld controllers. Both types scored above 7.5 (Table 2), indicating high acceptance by older adults. Initially worried about their ability to handle the high-tech equipment, after the actual experiment, they felt they were not as worried as they were; instead, many of them were glad to share their virtual experience with their younger family.

Notably, the asymmetric controller was rated significantly better for trigger controllability ($P = 0.017$). The most significant advantage of asymmetric board control is that seniors use the trigger control to pick up things by doing what feels natural. With two triggers (index and middle finger), its design mirrors natural hand use, allowing intuitive,

Table 1

The relationship between the participants' actions and the handheld controller.

Usability of the handheld controller	Actions of the participants
1. Trigger controllability	I.3, II.3
2. Directional controllability	I.2, II.2, II.6
3. Functional controllability	II.4, II.7
4. Mapping controllability	All actions with handheld controller in the game

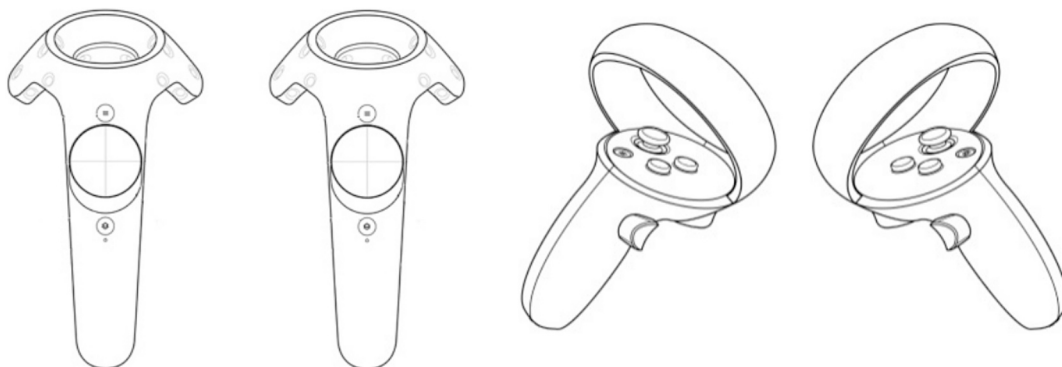


Fig. 1. Symmetric and asymmetric controllers.

Table 2
Usability of the handheld controller.

Handheld controller		Mean	SD	T-test (P-value)
1. Trigger controllability	symmetric	7.97	1.68	0.017*
	asymmetric	8.71	1.18	
2. Directional controllability	symmetric	8.14	1.81	0.074
	asymmetric	8.71	1.16	
3. Functional controllability	symmetric	8.40	1.78	0.235
	asymmetric	8.80	1.33	
4. Mapping controllability	symmetric	8.23	1.76	0.034*
	asymmetric	8.94	1.12	
5. Comfort	symmetric	8.20	2.00	0.317
	asymmetric	8.60	1.44	
6. Lightness	symmetric	7.71	1.86	0.007**
	asymmetric	8.80	1.33	
7. Enjoyment	symmetric	8.57	1.50	0.481
	asymmetric	8.74	1.46	

* P-value < 0.05; ** P-value < 0.01.

flexible control. Participants easily performed tasks like picking up objects using natural movements. In contrast, lacking ergonomic adaptation, the symmetric controller was often described as less intuitive – more like holding and firing a gun – requiring extra mental effort. The asymmetric design's alignment with natural hand posture greatly benefited senior users by reducing learning barriers and enhancing interaction fluency.

Regarding the mapping controllability of the handheld controller, most elderly participants considered the asymmetric mapping controllability to be significantly better than the symmetric one (P-value = 0.034; $P < 0.05$). The asymmetric grip fits the hand position very well, especially since the thumb can be easily positioned on the upper button area, and the thumb movement to the thumbstick and the A/B (X/Y) button can be easily switched into place. The index finger is only lightly put on the front trigger; the middle finger, ring finger, and pinky finger are wrapped around the grip, while the middle finger is just lightly placed on the other trigger. During the experiment, the whole palm and fingers are easily attached to the hand-mimicking controller and operate freely to touch the robot dog or pick up or throw the stick.

Regarding the lightness of the handheld controller, most elderly participants considered the asymmetric type to be significantly better than the symmetric one (P-value = 0.007; $P < 0.01$). In fact, the asymmetric Meta Quest 2 handheld controller (126 g) was lighter, while the symmetric HTC Vive handheld controller (200 g) was slightly heavier. The 35 participants seriously and consciously completed the questionnaire after experiencing the VR games. Some responded that the white Meta Quest 2 handheld controllers look lighter than the black HTC Vive in terms of visual design.

Concerning directional controllability, functional controllability, and comfort among the elderly participants in this experiment, we found no significant differences between the symmetric and asymmetric types of handheld controllers. Regarding directional controllability, older people did not feel any difference between the symmetric touchpad or asymmetric thumbstick operations. However, during the experiment, the researcher observed that they used the touchpad or thumbstick less for the direction of the virtual interaction and instead directly turned their body for interaction. As to whether or not such an action caused a motion sickness phenomenon afterwards, the results of the motion sickness questionnaire evaluation at least confirmed the existence of a correlation. Regarding functional controllability, there was no significant difference between older people's operation of the symmetric or asymmetric functions, as the average score was above 8. It is evident that they tend to be highly satisfied with the interaction between the VR device and the controller, and almost every participant showed interest and enjoyment in the virtual manipulation, with no one responding with or complaining about difficult operations. Finally, regarding comfort and enjoyment, there was no significant difference between symmetric and asymmetric controller operations on the comfort and enjoyment of

older people playing VR games. The average comfort score exceeds 8 and the average enjoyment score is also higher than 8 (shown in Table 2). This indicates that older people find the physical operations and visual experience of this VR game acceptable and do not experience discomfort; in fact, they find it enjoyable and pleasant. Older people have positive feedback about VR games and the controllers, demonstrating the potential of VR games to enhance the quality of life and entertainment options for the elderly.

4.2. Analysis of elderly VR sickness

After two 10-minute VR game sessions, each senior participant completed the Simulator Sickness Questionnaire (SSQ). The Postcards game allowed both short-distance walking and teleportation and required frequent body movements such as walking, turning, squatting, bending, grabbing, and throwing while interacting with a robot dog. These actions aimed to enhance immersion and help assess motion sickness symptoms.

Before the experiment, all 35 participants completed the SSQ pre-test. Four reported mild discomfort: one felt warm and sweaty, two were anxious about using the equipment, and one felt sleepy in the afternoon. These symptoms are related to oculomotor disturbance and nausea, but none reported orientation disorder issues. Despite mild pre-existing symptoms, all four still chose to participate. Thus, 31 participants started with no symptoms and four with mild ones. Details are shown in Tables 3 and 4.

After the first 10-minute experiment, of the four participants who initially had mild motion sickness, three remained mildly sick, while one reported worsening symptoms. Among the 31 participants who were normal at the start, 11 developed mild motion sickness ($11/31 = 35.48\%$), while the other 20 ($20/31 = 64.52\%$) remained unaffected. After a 10-minute break, all participants took part in a second 10-minute VR game. Of the original 31 normal participants, 18 stayed motion sickness-free throughout both experiments ($18/31 = 58.06\%$). Two additional participants developed symptoms, bringing the total to 13 with motion sickness ($13/31 = 41.94\%$). Among the 20 who had no symptoms in the first session, 18 remained symptom-free in the second ($18/20 = 90\%$).

This study did not apply the zero-baseline principle but followed Brown et al. [76] by using pre- and post-experiment SSQ assessments. Therefore, the statistics include the four participants with pre-existing mild motion sickness. The pre-test SSQ raw mean value was 0.171 (see Table 4).

After the first VR session, all 35 participants completed the SSQ. The average SSQ score was 2.514 (Table 4), indicating mild motion sickness. 20 participants reported no symptoms of nausea, oculomotor disturbance, and disorientation. The remaining 15 reported symptoms (7 for nausea, 14 for oculomotor disturbance, and 12 for disorientation).

After the second session, all 35 again completed the SSQ. The average score increased to 4.429 (Table 4). 18 reported no symptoms, while 17 showed various symptoms (11 for nausea, 16 for oculomotor disturbance, and 15 for disorientation).

It was clear that less than half of the older people experienced motion sickness after the VR play: 15 in the first post-test and 17 in the second (out of 35). The most obvious was the disorientation section, where no participants had orientation problems in the pre-test. However, the post-test showed that the weighted average of disorientation was much higher than the other two sections (Fig. 2). The T-test showed that the VR game made some older people feel dizzy, unstable, and disoriented, with difficulty focusing. This phenomenon requires further analysis through regression statistics.

After the first VR experiment, multiple regression analysis (Table 5) showed that all three predictors—Nausea, Oculomotor disturbance, and Disorientation—significantly affected the Total_Score ($p < 0.001$). The Oculomotor disturbance had the highest impact (Beta = 0.448), followed by Disorientation (Beta = 0.440), and Nausea had a smaller effect (Beta = 0.161). However, high VIF values for Oculomotor disturbance

Table 3

Count on the motion sickness pre-test and post-test for the elderly on VR games.

Motion Sickness Count														
SSQ items		Pre-Test				First Post-Test				Second Post-Test				-
		Never	Mild	Moderate	Severe	Never	Mild	Moderate	Severe	Never	Mild	Moderate	Severe	
Q1.	General discomfort	33	2	0	0	34	1	0	0	33	1	1	0	-
Q2.	Fatigue	34	1	0	0	32	3	0	0	32	2	1	0	
Q3.	Headache	35	0	0	0	33	2	0	0	33	1	1	0	
Q4.	Eyestrain	35	0	0	0	34	1	0	0	31	4	0	0	
Q5.	Difficulty focusing	35	0	0	0	28	3	3	1	24	5	5	1	
Q6.	Increased salivation	35	0	0	0	35	0	0	0	35	0	0	0	
Q7.	Sweating	34	1	0	0	35	0	0	0	35	0	0	0	
Q8.	Nausea	35	0	0	0	34	1	0	0	34	0	1	0	
Q9.	Difficulty concentrating	35	0	0	0	28	5	2	0	24	4	5	2	
Q10.	Fullness of head	35	0	0	0	34	1	0	0	34	1	0	0	
Q11.	Blurred vision	35	0	0	0	29	2	3	1	28	3	3	1	
Q12.	Dizzy (eyes open)	35	0	0	0	33	2	0	0	29	4	2	0	
Q13.	Dizzy (eyes closed)	35	0	0	0	33	2	0	0	27	6	2	0	
Q14.	Vertigo	35	0	0	0	27	7	1	0	25	5	5	0	
Q15.	Stomach awareness	35	0	0	0	35	0	0	0	35	0	0	0	
Q16.	Burping	35	0	0	0	35	0	0	0	35	0	0	0	

Table 4

Statistics on the motion sickness pre-test and post-test for the elderly on VR games.

SSQ items	Pre-Test		First Post-Test		Second Post-Test	
	Mean	SD	Mean	SD	Mean	SD
Q1. General discomfort	0.057	0.232	0.029	0.167	0.086	0.368
Q2. Fatigue	0.029	0.167	0.086	0.280	0.114	0.398
Q3. Headache	0	0	0.057	0.232	0.086	0.368
Q4. Eyestrain	0	0	0.029	0.167	0.114	0.318
Q5. Difficulty focusing	0	0	0.343	0.754	0.514	0.841
Q6. Increased salivation	0	0	0	0	0	0
Q7. Sweating	0.029	0.167	0	0	0	0
Q8. Nausea	0	0	0.029	0.167	0.057	0.333
Q9. Difficulty concentrating	0	0	0.257	0.553	0.571	0.935
Q10. Fullness of head	0	0	0.029	0.167	0.029	0.167
Q11. Blurred vision	0	0	0.314	0.747	0.343	0.754
Q12. Dizzy (eyes open)	0	0	0.057	0.232	0.229	0.539
Q13. Dizzy (eyes closed)	0	0	0.057	0.232	0.286	0.564
Q14. Vertigo	0	0	0.257	0.498	0.429	0.728
Q15. Stomach awareness	0	0	0	0	0	0
Q16. Burping	0	0	0	0	0	0
Nausea (Original)	0.086	0.280	0.314	0.708	0.714	1.343
Nausea (Weight)	0.818	2.671	2.998	6.754	6.814	12.814
Oculomotor disturbance (Original)	0.086	0.280	1.114	1.968	1.829	2.602
Oculomotor disturbance (Weight)	0.650	2.122	8.446	14.917	13.861	19.723
Disorientation (Original)	0	0	1.086	1.933	1.886	2.754
Disorientation (Weight)	0	0	15.113	26.904	26.249	38.342
Total Score (Original)	0.171	0.506	2.514	4.397	4.429	6.308
Total Score (Weight)	0.641	1.894	9.403	16.446	16.563	23.591

(16.381) and Disorientation (10.943) indicate serious multicollinearity, which may reduce regression model stability and clarity.

After the second VR experiment, multiple regression analysis (Table 6) showed all three predictors—Nausea_2nd, Oculomotor disturbance_2nd, and Disorientation_2nd—significantly affected

Total_Score_2nd ($p < 0.001$). Disorientation_2nd had the strongest effect (Beta = 0.437), followed by Oculomotor disturbance_2nd (Beta = 0.413), and Nausea_2nd (Beta = 0.213). VIF results show moderate to high multicollinearity for Oculomotor disturbance_2nd (VIF = 7.656), while Disorientation_2nd (VIF = 5.234) and Nausea_2nd (VIF = 2.854) are within acceptable ranges.

In both experiments, Oculomotor disturbance and Disorientation are the most influential predictors of Total_Score, while Nausea has a relatively smaller effect.

4.2.1. Nausea factors

Nausea levels showed no significant change between the pre-test and the first post-test ($P = 0.0882 > 0.05$) but a marginally significant increase between the pre-test and the second post-test ($0.05 > P = 0.0129 > 0.01$). A significant difference was found between the first and second post-tests ($P = 0.0062 < 0.01$) (Table 7). This indicates a slight but meaningful increase in nausea every 10 min after the experiment. Among the sub-items, Nausea and Difficulty Concentrating were identified as key factors.

Before the experiment, 2 participants reported mild malaise, 1 had mild sweating, and 32 reported no nausea (Table 3). After the first post-test, 5 reported mild nausea, 2 moderate, and 28 had no symptoms (Fig. 3). A regression analysis (Table 8) showed Q9 (Difficulty concentrating, Beta = 0.780) and Q8 (Nausea, Beta = 0.472) were significant predictors of nausea ($p < 0.001$); the other five items were not. Both had high tolerance (0.948) and low VIF (1.055), indicating no multicollinearity.

After the second post-test, 11 participants experienced difficulty concentrating: 2 had severe nausea, 5 moderate, and 4 mild; 24 reported no nausea (Fig. 3). A multiple regression analysis (Table 9) on seven predictors showed that only Q1_2nd (General discomfort), Q8_2nd (Nausea), and Q9_2nd (Difficulty concentrating) were significant ($p < 0.001$). The remaining predictors had no effect. Q9_2nd had the strongest impact (Beta = 0.696), followed by Q1_2nd (Beta = 0.274) and Q8_2nd (Beta = 0.248). All predictors had high tolerance (0.726–0.741) and low VIF (1.349–1.378), indicating no multicollinearity.

We can thus infer that VR games are less likely to produce immediate digestive nausea in older people (Table 3, Fig. 2). One-third of the participants felt mild Difficulty Concentrating after undergoing two experiments, and this issue of Q9 Difficulty Concentrating also affected the Oculomotor disturbance factor.

4.2.2. Oculomotor disturbance factors

We found significant difference between pre-test and first post-test,

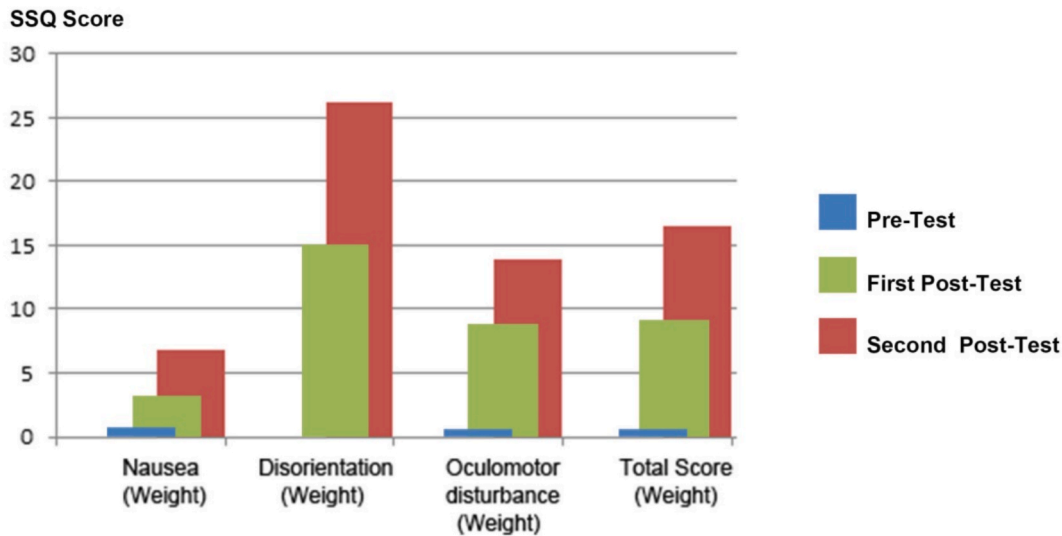


Fig. 2. SSQ score (weight) in the pre-test although the post-test.

Table 5

Multiple regression analysis of Nausea, oculomotor disturbance, and disorientation after the first VR experiment.

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error				Tolerance	VIF
1	(Constant)	−8.882E-16	0.000		0.000	1.000		
	Nausea	0.392	0.000	0.161	62213123.188	0< .001	0.309	3.235
	Oculomotor	0.493	0.000	0.448	76847037.018	0< .001	0.061	16.381
–	Disorientation	0.269	0.000	0.440	92338732.296	0< .001	0.091	10.943

a. Dependent Variable: Total_Score.

Table 6

Multiple regression analysis of Nausea, oculomotor disturbance, and disorientation after the second VR experiment.

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error				Tolerance	VIF
1	(Constant)	−5.329E-15	0.000		0.000	1.000		
	Nausea_2nd	0.392	0.000	0.213	25637204.857	0< .001	0.350	2.854
	Oculomotor_2nd	0.493	0.000	0.413	30321364.416	0< .001	0.131	7.656
	Disorientation_2nd	0.269	0.000	0.437	38821262.414	0< .001	0.191	5.234

a. Dependent Variable: Total_Score_2nd.

Table 7

T-test on the motion sickness pre-test and post-test for the elderly on VR games.

	Pre-Test & First Post-Test	Pre-Test & Second Post-Test	First Post-Test & Second Post-Test
Item	T-test (P-value)	T-test (P-value)	T-test (P-value)
Nausea	0.0882	0.0129	0.0062**
Oculomotor disturbance	0.0036**	0.0003**	0.0007**
Disorientation	0.0024**	0.0003**	0.0006**
Total	0.0037**	0.0004**	0.0003**

** P-value < 0.01.

significant difference between pre-test and second post-test, and significant difference between first post-test and second post-test (Table 7). Initially, 2 participants experienced mild uncomfortable, and 1 participant experienced mild sleepy in the oculomotor disturbance pre-test (Table 3); 32 participants reported feeling no oculomotor disturbance.

After the first post-test, 9 participants reported mild oculomotor disturbance, 3 moderate, and 2 severe; 21 reported no symptoms

(Fig. 4). A multiple regression analysis (Table 10) showed that among seven predictors, Q5 (Difficulty focusing, Beta = 0.384) and Q11 (Blurred vision, Beta = 0.381) had the strongest effects, followed by Q9 (Difficulty concentrating, Beta = 0.281). The other predictors (Q1–Q4) had smaller but still significant effects (Beta = 0.085–0.143, $p < 0.001$). However, multicollinearity was found in Q5, Q9, and Q11, with low tolerance (0.173–0.212) and high VIF values (4.711–5.774), indicating

Number of participant for Nausea

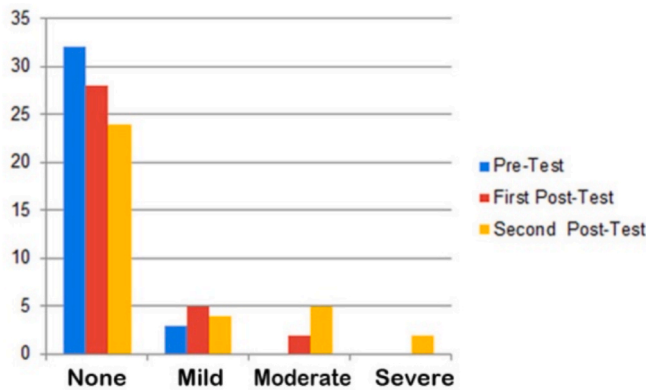


Fig. 3. Number of participant for nausea.

strong intercorrelation and reduced model stability.

After the second post-test, 16 participants reported oculomotor disturbance: 3 severe, 7 moderate, and 6 mild; 19 reported no symptoms (Fig. 4). A regression analysis (Table 11) found that Q9_2nd (Difficulty concentrating, Beta = 0.359) and Q5_2nd (Difficulty focusing, Beta = 0.323) were the strongest predictors, followed by Q11_2nd (Blurred vision, Beta = 0.290). The other predictors (Q1–Q4_2nd) had smaller but significant effects (Beta = 0.122–0.153). Multicollinearity was present in Q5_2nd, Q9_2nd, and Q11_2nd, with low tolerance (0.213–0.295) and high VIF values (3.393–4.699). In contrast, Q1_2nd–Q4_2nd showed no collinearity issues.

Based on the above two regression analyses, we can infer that older people who experience oculomotor disturbance while using VR are more likely to exhibit motion sickness symptoms, specifically difficulty concentrating, difficulty focusing, or blurred vision.

4.2.3. Disorientation factors

We found significant difference between pre-test and first post-test, significant difference between pre-test and second post-test, significant difference between first post-test and second post-test (Table 7). No participants had symptoms of disorientation in the pre-test (Table 3), all being very healthy concerning the disorientation section.

After the first post-test, 7 participants reported mild disorientation, 3 moderate, and 2 severe; 23 had no symptoms (Fig. 5). A regression analysis (Table 12) showed that six of seven predictors were significant ($p < 0.001$). Q5 (Difficulty focusing, Beta = 0.390) and Q11 (Blurred vision, Beta = 0.387) had the strongest effects, followed by Q14 (Vertigo, Beta = 0.258) and Q13 (Dizzy (eyes closed), Beta = 0.240). Q8 (Nausea) and Q10 (Fullness of head) had smaller but notable effects (Beta = 0.086 each). Multicollinearity was moderate for Q5 and Q11, with low tolerance (0.246–0.257) and high VIFs (3.893–4.069), while the remaining predictors showed no collinearity issues.

After the second post-test, 2 participants reported severe disorientation, 6 moderate, and 7 mild; 20 had no symptoms (Fig. 5). A regression analysis (Table 13) found all seven predictors to be significant ($p < 0.001$). Q5_2nd (Difficulty focusing, Beta = 0.305) and Q11_2nd (Blurred vision, Beta = 0.274) were the most influential, followed by Q14_2nd (Vertigo, Beta = 0.264), Q13_2nd (Dizzy (eyes closed), Beta = 0.205), and Q12_2nd (Dizzy (eyes open), Beta = 0.196). Q8_2nd (Nausea) and Q10_2nd (Fullness of head) had smaller effects (Beta = 0.121 and 0.060). Multicollinearity was high for Q12_2nd and Q13_2nd, with very low tolerance (0.080–0.077) and high VIFs (12.475–13.057). The remaining predictors showed no major collinearity concerns.

Based on the above two regression analyses, we can thus infer that

Number of participant for Oculomotor disturbance

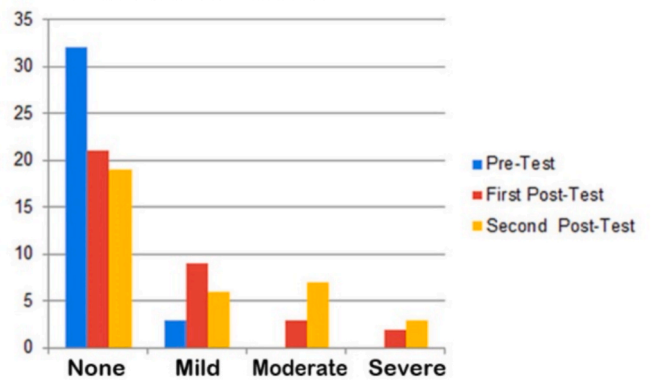


Fig. 4. Number of participant for oculomotor disturbance.

Table 8

Multiple regression analysis of seven predictors on Nausea after the first VR experiment.

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	1.110E-16	0.000		0.000	1.000		
	Q8	19.080	0.000	0.472	87173087.710	0<.001	0.948	1.055
	Q9	9.540	0.000	0.780	144161424.323	0<.001	0.948	1.055

a. Dependent Variable: Nausea.

Table 9

Multiple regression analysis of seven predictors on Nausea after the second VR experiment.

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	2.554E-15	0.000		0.000	1.000		
	Q1_2nd	9.540	0.000	0.274	42920992.888	0<.001	0.741	1.349
	Q8_2nd	9.540	0.000	0.248	38706978.773	0<.001	0.736	1.359
	Q9_2nd	9.540	0.000	0.696	107821481.536	0<.001	0.726	1.378

a. Dependent Variable: Nausea_2nd.

Table 10

Multiple regression analysis of seven predictors on Oculomotor Disturbance after the first VR experiment.

Coefficients ^a		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
Model		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	9.326E-15	0.000		0.000	1.000		
	Q1	7.580	0.000	0.085	8071911.612	0<.001	0.672	1.489
	Q2	7.580	0.000	0.143	16260863.927	0<.001	0.967	1.034
	Q3	7.580	0.000	0.118	11215950.660	0<.001	0.669	1.495
	Q4	7.580	0.000	0.085	9781025.265	0<.001	0.986	1.014
	Q5	7.580	0.000	0.384	20476488.655	0<.001	0.212	4.711
	Q9	7.580	0.000	0.281	13556401.308	0<.001	0.173	5.774
	Q11	7.580	0.000	0.381	18847637.874	0<.001	0.183	5.469

a. Dependent Variable: Oculomotor.

Table 11

Multiple regression analysis of seven predictors on Oculomotor Disturbance after the second VR experiment.

Coefficients ^a		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
Model		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	8.882E-15	0.000		.	.		
	Q1_2nd	7.580	0.000	0.141	.	.	0.460	2.173
	Q2_2nd	7.580	0.000	0.153	.	.	0.751	1.332
	Q3_2nd	7.580	0.000	0.141	.	.	0.849	1.178
	Q4_2nd	7.580	0.000	0.122	.	.	0.603	1.658
	Q5_2nd	7.580	0.000	0.323	.	.	0.295	3.393
	Q9_2nd	7.580	0.000	0.359	.	.	0.290	3.448
–	Q11_2nd	7.580	0.000	0.290	.	.	0.213	4.699

a. Dependent Variable: Oculomotor_2nd.

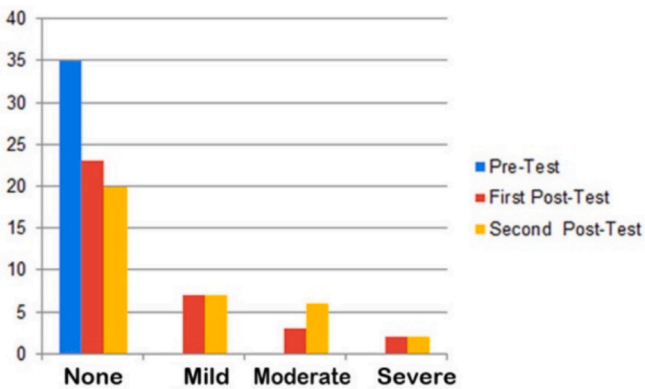
Number of participant for Disorientation

Fig. 5. Number of participant for disorientation.

when older people use VR and experience disorientation, they are most likely to exhibit symptoms such as difficulty focusing or blurred vision.

4.2.4. Key takeaways

Regarding oculomotor disturbance and disorientation issues, there was a significant difference between the pre-test and post-test. Some seniors responded with somewhat blurred vision and some experienced moderately severe dizziness and inability to focus. From the regression analyses, it can be summarized that when older people use VR, they are more likely to experience difficulty focusing or blurred vision. These two signs, in turn, contribute to both oculomotor disturbance and disorientation, as they are predictors of both symptoms. This phenomenon highlights the interconnected nature of these symptoms and their influence on the overall VR experience for older people.

Regarding nausea issues, no cases of digestive nausea reported; only a few more participants than the pre-test responded with slight body uneasiness and discomfort. In short, more than half of the older people did not react after two experiments. In both experiments, we found that the seniors prone to dizziness developed mild dizziness in the first 10-minute experiment, and their dizziness increased after the second

Table 12

Multiple regression analysis of seven predictors on Disorientation after the first VR experiment.

Coefficients ^a		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
Model		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	–2.665E-15	0.000		0.000	1.000		
	Q5	13.920	0.000	0.390	39032309.320	0<.001	0.246	4.069
	Q8	13.920	0.000	0.086	15848632.521	0<.001	0.829	1.206
	Q10	13.920	0.000	0.086	17312107.982	0<.001	0.990	1.010
	Q11	13.920	0.000	0.387	39559084.957	0<.001	0.257	3.893
	Q13	27.840	0.000	0.240	37529680.583	0<.001	0.599	1.670
–	Q14	13.920	0.000	0.258	39721323.854	0<.001	0.583	1.716

a. Dependent Variable: Disorientation

Table 13

Multiple regression analysis of seven predictors on Disorientation after the second VR experiment.

Coefficients ^a		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
Model		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	1.465E-14	0.000		0.000	1.000		
	Q5_2nd	13.920	0.000	0.305	44648800.403	0<.001	0.329	3.041
	Q8_2nd	13.920	0.000	0.121	29268014.428	0<.001	0.900	1.111
	Q10_2nd	13.920	0.000	0.060	15318372.572	0<.001	0.986	1.014
	Q11_2nd	13.920	0.000	0.274	36848815.039	0<.001	0.279	3.588
	Q12_2nd	13.920	0.000	0.196	14133529.402	0<.001	0.080	12.475
	Q13_2nd	13.920	0.000	0.205	14459588.074	0<.001	0.077	13.057
–	Q14_2nd	13.920	0.000	0.264	35023293.418	0<.001	0.270	3.709

a. Dependent Variable: Disorientation_2nd.

experiment. However, most seniors (90 %) who did not experience dizziness in the first 10-minute experiment did not experience dizziness in the second experiment. This result is a promising finding. We may presume that over half of the older people are suitable for VR games to supplement their physical and mental health.

A comparison of SSQ scores between the two controller types revealed no statistically significant differences in nausea, oculomotor disturbance, or disorientation. However, qualitative observations indicated indirect effects related to controller design. A small number of older adults turned their bodies more frequently when using symmetrical controllers, whereas with asymmetrical controllers, they focused more on the thumbsticks or touchpads. This increased body movement may have led to increased disorientation, although the effect was not statistically significant. In contrast, the ergonomic design of the asymmetrical controller appeared to support more intuitive interactions and reduce unnecessary movements.

5. Discussion

Our study revealed that most older participants were eager to learn how to play VR games and enjoyed the experience. Although their responses on the Simulator Sickness Scale (SSQ) and handheld controller usability questionnaires indicated some issues with experiencing VR games, they later told researchers they were happy to participate and excited to share their experiences with relatives and friends. So, these issues could be addressed in future developments. Notably, their limited access to techno-electronic devices did not affect their ability to feel comfortable in virtual environments, experience emotions similar to real-life situations, enjoy interactions, or perform VR tasks.

We investigated the impact of immersion and motion sickness on older people, focusing on healthy, mobile, and energetic volunteers aged 70 and above. According to our study, the situation has improved among senior users, aligning with claims by equipment providers about advancements in next-generation controllers and HMDs. Inconsistencies between visual and proprioceptive senses caused motion sickness, disorientation, and nausea in just under half of the participants. While some previous studies excluded participants who withdrew due to discomfort to protect their health, this practice reduces the accuracy of motion sickness assessments. By including only physically capable seniors, our study provides a clearer understanding of the issue.

Therefore, our study employed a two-stage experiment to assess motion sickness in older people during VR gameplay. Participants were given a 10-minute break between the first and second sessions to compare motion sickness symptoms across pre-test and post-test stages, using a staged validation. Of the 35 participants, 20 (57.14 %) showed no signs of motion sickness after the first session, and 18 (51.43 %) remained symptom-free after the second. Adjusting for the four participants who experienced motion sickness prior to the experiment, the proportions rose to 64.52 % and 58.06 %, respectively. These findings suggest older people, in general, tolerate VR sickness better than

younger people, consistent with previous research. Furthermore, older people without motion sickness enjoyed VR games as a means of physical and intellectual engagement, with opportunities to interact socially with family members.

Conversely, 17 participants reported varying levels of motion sickness on the Simulator Sickness Questionnaire (SSQ), contributing to the overall scores. Weighted comparisons of nausea, oculomotor disturbance, and disorientation before and after the sessions revealed significant increases in symptoms, aligning with prior studies that older people experience increased motion sickness due to longer experimental time. For instance, Park et al. [30] observed higher dropout rates due to VR sickness among older adults (70–90 years) than younger groups. Similarly, Moss and Muth [78] found that VR sickness symptoms worsened with exposure by 2-minute increments, peaking in the final 10 min. Our findings echo these patterns, with motion sickness increasing significantly in the second 10-minute session, demonstrating that prolonged VR exposure exacerbates symptoms in older users.

That being said, older people aged 70 and above demonstrated greater endurance to VR sickness compared to younger individuals. Our study yielded the following mean SSQ scores: for the first experiment, Nausea = 2.998, Oculomotor = 8.446, Disorientation = 15.113, and Total Score = 9.403; for the second experiment, Nausea = 6.814, Oculomotor = 13.861, Disorientation = 26.249, and Total Score = 16.563. These results highlight notable differences from previous studies. Saredakis et al.' review (2020) of 55 studies on VR sickness reported total SSQ scores ranging from 14.30 to 35.27, with an average participant age of 24 years. In comparison, our study's first experiment recorded a significantly lower Total Score of 9.403 (just over half of the minimum score in the range), while the second experiment's Total Score of 16.563 was slightly above the lower bound. Notably, most of the reviewed studies focused on younger participants, with 45 out of 54 papers indicating an average age of 24 years. Comparing our results with these studies, we found that older adults exhibited much lower motion sickness scores than younger individuals. This difference might partially stem from variations in experimental conditions. While Saredakis et al. did not explicitly state that susceptibility to VR sickness decreases with age, they observed significantly lower mean SSQ scores among older participants (average age ≥ 35 years) compared to younger ones. In this respect, our study further enhances the thesis that the older the VR users are, the lower the SSQ scores. Our study showed that older people aged 70 and above are much better off than young people regarding VR sickness.

Our study also examined the impact of symmetric versus asymmetric handheld controllers on older participants' gameplay. While both types of controllers have their advocates among younger gamers, older participants in our study – who were all VR novices without prior exposure to such devices – responded more positively to the intuitive design of hand-mimicking asymmetric controllers. When using asymmetric controllers, they could naturally perform gross actions like grabbing, releasing, and throwing with minimal learning effort, thanks to the

trigger's design that closely mimics real-life hand movements. The experiment took place in a fully immersive outdoor wilderness setting, where participants engaged in high levels of interaction with a virtual robot dog. This environment enhanced their sense of interactive immersion, making the experience enjoyable and relatable. Interacting with the virtual dog in ways resembling everyday activities further helped older people adapt to the handheld controllers with greater ease.

Some older people initially expressed apprehension about high-tech devices, fearing they might be difficult to use, fragile, or overwhelming to learn. However, the VR experience demonstrated that if devices are designed to look and operate naturally, older users are not only willing to try them but also find satisfaction in engaging with new technology. For instance, asking older novice players to pick up objects using instinctive motions, rather than perceiving the device as a controller, created a significant advantage in usability and mapping. This highlights the need to develop handheld controllers and VR equipment that prioritize intuitive design, tailored to the needs of older users. Addressing these usability concerns from a user-experience perspective could make VR systems more accessible and enjoyable for this demographic, paving the way for broader adoption.

Despite these valuable insights, this research has several limitations that warrant acknowledgment. First, the sample size was relatively small and restricted to Taiwanese seniors, potentially limiting the generalizability of the findings to other cultural or geographical contexts. Future studies should consider recruiting larger and more diverse participant groups to investigate the influence of cultural background on VR use among older adults. Second, the study relied on the Simulator Sickness Questionnaire (SSQ) to assess participants' motion sickness. While the SSQ is widely used, its application among older adults presents certain challenges, particularly for individuals with cognitive limitations, which could affect the accuracy of their responses. Future research should consider integrating physiological measures, such as heart rate or galvanic skin response, to provide more objective assessments of motion sickness. In terms of experimental design, the VR sessions were limited to two 10-minute segments with a 10-minute rest interval in between to accommodate the physical endurance of older participants. However, this design may not adequately capture the effects of prolonged VR use on older adults. Future research should extend the duration of VR exposure to better understand the long-term impacts. Finally, the study was limited by the types of VR equipment and content utilized. Only two commonly available handheld controllers and a single VR game were used as the test environment. Consequently, the findings may not be applicable to other types of VR devices or applications, such as those employed in rehabilitation settings. Future studies should explore a wider range of VR equipment and content to examine user experiences across diverse application scenarios.

6. Conclusion

In conclusion, this study demonstrates that Taiwanese older adults can effectively use VR devices and enjoy the experience, even if they initially express apprehension toward high-tech products. The findings highlight that older people aged 70 and above exhibit considerable resilience to VR-induced motion sickness compared to younger people, reinforcing previous research suggesting that susceptibility to VR sickness decreases with age.

The study also underscores the importance of intuitive design for VR controllers, particularly for older people with limited familiarity with technology. The asymmetric controller design proved to be more intuitive and user-friendly, allowing participants to interact with the VR environment naturally without significant learning efforts. The finding indicates that user-centered design improvements can enhance the accessibility and overall enjoyment of VR experiences for older populations.

Overall, Taiwanese older people are aware of the rapid advancements in technology. The inability to keep pace with the younger

generation may exacerbate the generational divide. An interesting finding from the present study is that over half of the senior participants did not experience motion sickness while using VR. In addition, they reported feeling comfortable in virtual environments and expressed enjoyment in completing VR tasks and interactions. Despite the physical challenges of aging, older people showed a willingness to engage with VR, particularly as they experienced fewer issues with motion sickness. Developing VR serious games tailored to older users could help them remain entertained and healthy in a sustainable manner. Although VR games have gradually gained popularity in recent years, motion sickness remains one of the major factors that hinder the development of the VR industry [79,80]. Since, according to our study and similar previous research findings, older adults are less prone to motion sickness, this demographic represents a niche market for VR serious games, particularly for health, in a future super-aged society. For example, rehabilitation therapy is often a long and monotonous process, requiring consistent effort to achieve effective results. However, VR can enhance the enjoyment and engagement of rehabilitation, making clinical patients more willing to participate actively in their therapy. Building upon this technological foundation, further collaboration with medical professionals can facilitate the development of more interactive rehabilitation training programs.

We have included specific recommendations for future research to understand better and enhance the role of VR technology for seniors. One important area for exploration is conducting longitudinal studies to examine the effects of sustained interaction with VR on seniors' health, usability, and willingness to engage with such technology. Additionally, further research could explore the impact of different physiological factors, such as age, height, weight, and chronic illnesses, on the VR experience. Such longer-term studies would enable researchers to track changes in user satisfaction, ease of use, and measurable health outcomes over time, providing a more comprehensive understanding of VR's potential impact. Another critical aspect is developing and evaluating diversified VR content tailored to the needs and interests of older adults. Future research should investigate how different types of applications, such as virtual tourism, exercise games, cognitive exercises, social interaction simulations, and immersive rehabilitation programs, impact senior engagement, deliver specific benefits, and influence motion sickness.

In addition, structured and content gamification strategies—originally developed to enhance learning performance and engagement—show promise for motivating older people in VR environments. Takbiri et al. [81] highlighted the effectiveness of psychologically grounded gamification systems that incorporate progression levels, rewards, and narrative structures to sustain user motivation. Applying similar frameworks in health-oriented VR applications may enhance older adults' motivation, participation, and long-term adherence, particularly in rehabilitation programs requiring repetitive engagement. Extending this approach, Das et al. [82] conducted a systematic review and proposed that spatial gamification mechanisms—those leveraging spatial cues, location-based prompts, and embodied interaction—can significantly enhance immersion, engagement, and embodied learning in digital learning environments. For older users, who often experience spatial disorientation in virtual settings, these spatially grounded gamification strategies may offer not only motivational benefits but also essential cognitive support. Our study directly utilized existing game content to allow older people to experience and explore their use of VR. Future research can apply structured or content gamification frameworks to customize VR content for older people, thereby enhancing motivation and engagement. Such approaches may be more effectively applied in VR experiences related to learning, rehabilitation, and muscle training for seniors. Furthermore, future research may explore whether structured and content-based gamification strategies effectively encourage older adults to participate in serious VR games and how spatially structured gamification can mitigate orientation-related challenges to provide a more intuitive, accessible VR experience for older

users.

Finally, integrating VR technology into healthcare settings holds significant promise for practical applications. Collaboration between technology developers and healthcare providers could ensure that VR based serious games are designed to meet the specific needs of seniors, particularly in rehabilitation and health maintenance. By pursuing these approaches, future research and development efforts can more effectively harness the potential of VR technology to support the well-being and quality of life of older adults.

CRediT authorship contribution statement

Cheih-Ying Chen: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Appendix. SSQ Computation

Nausea	Oculomotor disturbance	Disorientation
1.General discomfort 6.Increased salivation 7.Sweating 8.Nausea 9.Difficulty concentrating 15.Stomach awareness 16.Burping	1.General discomfort 2.Fatigue 3.Headache 4.Eyestrain 5.Difficulty focusing 9..Difficulty concentrating 11.Blurred vision	5.Difficulty focusing 8.Nausea 10.Fullness of head 11.Blurred vision 12.Dizzy (eyes open) 13.Dizzy (eyes closed) 14.Vertigo
Total nausea symptom judgments= $\sum N$	Total oculomoto symptom judgments= $\sum N$	Total disorientation symptom judgments= $\sum N$
Nausea symptom weighted score [N] = $\sum N \times 9.54$ (score range: 0–200.34)	Oculomoto symptom weighted score [O] = $\sum N \times 7.58$ (score range: 0–292.32)	Disorientation symptom weighted score [D] = $\sum N \times 13.92$ (score range: 0–159.18)
SSQ Total Score [TS]=[N] + [O] + [D] x 3.74 (score range: 0–179.52)		

Data availability

Data will be made available on request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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